

Slovak University of Technology in Bratislava
Faculty of Informatics and Information Technologies

FIIT-XXXX-XXXXX

Bc. Richard Čerňanský

**Deep Learning Methods for Perception and
Prediction in Autonomous Systems**

Master's Thesis

Thesis Supervisor: Ing. Matej Halinkovič, PhD.

May 20xx

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Study Programme:	Intelligent Software Systems
Study Field:	Computer Science
Training Workplace:	Bratislava, Slovakia
Thesis Supervisor:	Ing. Matej Halinkovič, PhD.

May 20xx

Čestne vyhlasujem, že som túto prácu vypracoval samostatne, na základe konzultácií a s použitím uvedenej literatúry.

I declare in my honor that I have prepared this work independently, on the basis of consultations and using the mentioned literature.

In Bratislava, May 20xx

Bc. Richard Čerňanský

Annotation

Slovak University of Technology in Bratislava
Faculty of Informatics and Information Technologies
Study Programme: Intelligent Software Systems
Study Field: Computer Science

Author: Bc. Richard Čerňanský
Master's Thesis: Deep Learning Methods for Perception and Prediction in Autonomous Systems
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May 20xx

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Keywords: 3-5 keywords

Anotácia

Slovenská technická univerzita v Bratislave
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Diplomová práca: Deep Learning Methods for Perception and Prediction in Autonomous Systems

Vedúci diplomového projektu: Ing. Matej Halinkovič, PhD.

Máj 20xx

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Oblasť problematiky: 3-5 kľúčových slov

BP2, DP3: sem vložte finalne a podpísane zadanie: PDF z AISu

BP1, DP1, DP2: sem vložte návrh zadania: PDF z YonBanu

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Chapter 1

Introduction

In recent years, deep learning and computer vision methods have made significant advancements in the analysis of dynamic scenes and their applications in autonomous systems. The development of fields such as autonomous vehicles and robotics depends on solving tasks like object detection, object tracking, segmentation, and trajectory prediction. Specifically, autonomous vehicles (AVs) revolutionize transportation by saving drivers' time, improving safety and possibly mitigating congestion on the roads. Current autonomous driving systems consist of multiple sequential tasks such as perception, tracking, prediction, planning and control [1]. Perception task is responsible for detecting and understanding the objects within the environment including their categorization, location and size estimation. The tracking objective uses the detections from perception to find the relationships between objects across the frames in scene. The prediction modules then process this tracking information in order to forecast future positions. The predicted positions enter the planning stage that generates the most feasible and optimal path in terms of safety, energy consumption and following traffic rules. Last but not least, controlling task then prepares actuator commands which execute the planned motion.

1.1 Motivation

Real traffic is highly dynamic and unpredictable: other drivers may behave erratically, and sensors provide noisy or incomplete data. Diverse maneuvers, complex interactions, and inherent errors in sensory information make prediction challenging, yet solving these challenges is essential for having trustworthy AVs. In addition, autonomous systems operate under strict constraints on latency and reliability [2]. Predictions must be computed in real time on embedded hardware and remain accurate over long hori-

zons to be useful for driving in real-life scenarios. Industry recognizes the need for fast execution, uncertainty handling, and generalization in prediction models to ensure robust operation in varied conditions [3, 4, 5]. Reliable trajectory prediction is therefore critical for safe autonomous driving and motivates intensive research into methods that can forecast the multi-agent future with practical efficiency. For these reasons, this thesis focuses on trajectory prediction, exploring how current state-of-the-art models could be modified or extended to achieve better accuracy and efficiency.

1.2 Problem Formulation

Trajectory prediction is the task of forecasting the future motion of the traffic participants that surround the ego vehicle. Its input is not a trajectory in the purely geometric sense, but rather the past states of these participants as they are captured in the environmental information collected by sensors, such as camera images, LiDAR point clouds, radar returns or HD maps providing lane geometry and traffic-rule priors. However, in this thesis only camera and LiDAR is employed. A model that relies on a single source is unimodal, whereas a model that exploits several of them at once is considered multi-modal. In the latter case the measurements are heterogeneous in dimensionality, sampling rate and coordinate frame, so the features extracted from the individual sources have to be fused into a unified representation of the scene before any forecasting takes place [6]. The output is the forecasted trajectory, that is, a sequence of future states of the observed agents over a fixed prediction horizon.

Formally, trajectory prediction can be stated as the problem of constructing a mapping Φ from the observed evidence $\mathcal{I}$ about the scene to the future states $\mathcal{O}$ of the traffic participants, from the current time up to a prediction horizon T [7]:

$$\begin{aligned}\Phi : \mathcal{I} &\rightarrow \mathcal{O}, \\ \mathcal{O} &= \{S^0, S^1, \dots, S^T\},\end{aligned}\tag{1.1}$$

where S^0 is the current state and S^1 to S^T are the states at the individual future time steps, each of them collecting the states of all N observed participants. Depending on the available modalities and on the downstream task, a single state can take different forms, for example

$$S^t = \{x^t, y^t\} \quad \text{or} \quad S^t = \{x^t, y^t, l^t, w^t, \theta^t\},\tag{1.2}$$

where x^t and y^t are the coordinates of the centre point of the object, l^t and w^t its length and width, and θ^t its orientation [7]. In trajectory prediction the state is usually reduced to the planar position, so that $\mathcal{O}$ corresponds to the set of forecasted trajectories of the

participants in the scene. Since the future of a traffic scene is inherently uncertain, the mapping in (1.1) is in practice often extended to output several plausible trajectories together with their confidences, allowing the planning stage to reason over the alternatives instead of committing to a single hypothesis [2]. The following subsections summarise the main challenges that this formulation raises.

1.2.1 Data Fusion

The information describing the scene comes from sources that are complementary but incompatible. Cameras provide dense semantic detail without depth and degrade at night, in glare or in poor lighting conditions, LiDAR provides accurate geometry but suffers in rain etc. These sources can be combined at the level of raw measurements (early), of learned features (intermediate) or of the individual predictions (late). Since every added modality costs latency and introduces noise the network is prone to overloading with parameters or can overspecialize on the most informative source and lose accuracy once the weather or illumination degrades it [6].

1.2.2 Dynamic Environments

Traffic is nonstationary, as density, weather, illumination and road layout change, so that the conditions met during deployment need not resemble those seen during development. The future is moreover not unique, since the same observed history is usually compatible with several maneuvers, and this aleatoric uncertainty originated in the erratic decisions of other drivers does not diminish with additional data. Therefore the task requires a set of socially compliant trajectories produced in time for the planning stage [6].

1.2.3 Interaction Modeling

The motions of traffic participants are not independent and traffic signals constrain all of them, so that the joint future does not split into separate per-agent distributions. Capturing the agent to agent and the agent to map relations requires representations able to operate on a variable and unordered number of entities, which is why structured formulations such as graphs and attention mechanisms prevail. Where these relations are modelled poorly, forecasts that appear plausible for each agent in isolation become inconsistent at the level of the scene comprising of multiple entities [2].

1.2.4 Real-World Constraints

Prediction is one stage of a control loop closing at a fixed rate, which leaves it a budget of tens of milliseconds on embedded hardware limited in memory and power. Its cost

grows with the number of participants and becomes quadratic once their interactions are modelled, so the least favourable case is dense traffic, precisely where prediction matters most. Model capacity therefore has to be weighed against computational efficiency, as accuracy alone does not make a model deployable [4, 2].

1.3 Thesis Outline

Chapter 2

Trajectory Prediction: Background and State of the Art

concrete header for the given thesis's theme!

2.1 Taxonomy of Trajectory Prediction Methods

In the literature, there are numerous different prediction methods, and several criteria enable their systematic classification, such as the modality and the representation of the input, the form of the produced output, or the paradigm by which prediction is integrated with perception. Here the modeling methods are classified according to the categorization presented in [6].

2.1.1 Physics-Based Methods

These rely on kinematic or dynamic models of motion, using physical equations to extrapolate future positions and not accounting for uncertainty. Examples include constant-velocity, constant-acceleration models (Single-Trajectory simulation) [8] [9]. Kalman filters address the uncertainty limitation of the previous models by modeling the vehicle's state as a Gaussian distribution. It predicts trajectories as a two step process of filtering (denoising current state from noisy measurements) and forecasting [10]. Monte Carlo methods, in particular particle filtering, drop the linearity and Gaussianity assumptions of the Kalman filter by approximating the state distribution with a set of weighted samples propagated through the motion model, which permits arbitrarily shaped and multimodal distributions at the cost of a higher computational load [11]. Physics-based approaches are simple and fast, but they struggle with complex maneuvers or

sudden behavior changes since they lack contextual understanding. Some variants combine multiple motion models to handle different maneuvers (Multiple-Model methods) [12].

2.1.2 Learning-based methods

Unlike physics-based, learning-based methods learn the motion dynamic from data. Here, we consider three groups of approaches.

2.1.2.1 Classical Machine Learning Methods

These methods learn motion patterns from data using statistical models, but without deep neural networks. Common approaches include hidden Markov models (HMMs) [13], Gaussian process (GPs) based prediction models [14], dynamic Bayesian networks (DBNs) [15], and Gaussian mixture models (GMMs) [16]. They can capture uncertainty to a degree (e.g., an HMM can model discrete maneuver states, GPs provide a distribution over trajectories) and were popular before deep learning. However, their capacity to model very complex, high-dimensional scenarios is limited compared to modern deep nets.

2.1.2.2 Reinforcement Learning Methods

Reinforcement learning (RL) methods treat prediction as a sequential decision problem. Here, an agent learns a policy (via reward feedback) to mimic human driving behavior, effectively planning likely trajectories. Since the reward that drives human driving is not known, inverse reinforcement learning (IRL) recovers it from expert demonstrations and uses it to regularise the predictor [17], while imitation learning (IL) skips the reward altogether and extracts the policy directly from the observed trajectories [18]. RL-based predictors can naturally handle multi-agent interactions by modeling how agents might react to each other, and can ensure physically feasible outputs through learned policy constraints. However, RL methods can be data-hungry, harder to train and stabilize, and historically have been more common in planning than in pure trajectory forecasting. They are an active research area for capturing game-theoretic interactions and long-horizon planning in prediction.

2.1.2.3 Deep Learning Methods

2.2 Prediction Paradigms

2.2.1 Modular (Detect-Track-Predict)

2.2.2 End-to-End (Joint Learning)

2.3 Datasets

Research in trajectory prediction has been accelerated by large-scale autonomous driving datasets that provide rich multimodal sensor data and trajectory annotations. Some prominent datasets include:

- **KITTI (2012):** Early autonomous vehicle (AV) benchmark with stereo, light detection and ranging (LiDAR), and global positioning system (GPS)/inertial measurement unit (IMU), great for detection, tracking, and simultaneous localization and mapping (SLAM). It is small (150 scenes) and supports only short-horizon forecasting, so it's limited for learning social behavior. [19]
- **nuScenes (2020):** Large multimodal suite (6 cameras, 5 radars, 1 LiDAR, 360°) with 1,000 twenty-second scenes, 3D boxes, and high-definition maps (HD maps). It added a prediction challenge (≈ 6 s horizon) and is ideal for multi-sensor fusion and map-aware forecasting. [20]
- **Argoverse (2019):** Includes a Motion Forecasting set with 324k five-second scenarios and rich HD lane graphs (Miami/Pittsburgh). It popularized map-based prediction and uses ADE/FDE and MR; Argoverse 2 expands coverage. [21]
- **Waymo Open Motion (2021):** Very large-scale forecasting dataset (>100 k twenty-second scenarios at 10 Hz) with HD maps and many interactive scenes. It introduced mAP and MR for multi-modal sets and is a key benchmark for vectorized/transformer models. [3]
- **Lyft Level 5:** Large urban dataset with synchronized LiDAR/cameras and HD maps, similar in format to nuScenes. Widely used for 3D detection/tracking and baseline motion forecasting with map context. [22]
- **ApolloScape:** Chinese urban driving data covering detection, segmentation, tracking, and mapping. Useful for perception and localization; less standardized for forecasting than nuScenes/Argoverse. [23]

2.4 Evaluation Metrics

Across these benchmarks, common metrics to evaluate trajectory prediction include [6, 3]:

- **Average Displacement Error (ADE):** the average Euclidean distance between the predicted trajectory and the ground-truth trajectory over all time steps. It measures overall prediction accuracy.
- **Final Displacement Error (FDE):** the Euclidean distance between the predicted final position (at the prediction horizon, e.g., 3 s or 6 s) and the true final position. This emphasizes endpoint accuracy (did the model correctly predict where the agent ends up?).
- **minADE and minFDE (multi-modal prediction):** for models that output K possible futures per agent, these take the error of the closest prediction to ground truth (assume the model's best hypothesis). minADE/FDE reflect the best-case performance of a probabilistic predictor.
- **Miss Rate (MR):** usually defined as the fraction of cases where none of the K predicted trajectories is within a certain distance (e.g., 2 m) of the ground truth at the final time. A lower miss rate means the model almost always has at least one prediction near the true outcome, which is important for safety.
- **Mean Average Precision (mAP):** introduced in Waymo's benchmark, mAP treats trajectory prediction as a detection problem in trajectory space. It evaluates the precision-recall curve of the model's predicted set of trajectories against ground truth, taking into account both spatial tolerance and assigned confidence scores. It effectively rewards methods that produce a set of diverse predictions that cover the ground truth while avoiding too many implausible guesses.
- **Other metrics:** Negative Log-Likelihood (NLL) for probabilistic methods (measuring the quality of the predicted probability distribution), and occasionally collision rate or overlap metrics to ensure predicted trajectories are physically feasible (not crashing into others or violating road constraints).

Using these metrics in combination provides a comprehensive evaluation: ADE/FDE for accuracy, minADE/minFDE and MR for multimodal coverage, and mAP for overall usefulness of the prediction set. The literature often reports all of them to compare methods. For instance, a method might achieve lower ADE than another (better average accuracy), but if its miss rate is high it means it sometimes totally misses possible outcomes, a trade-off the proposal's approach must consider.

2.5 Related Work

This part is important in determining the objectives of the Thesis as we will state in Section 2.6 and in Chapter 3.

2.6 Summary and Starting Points

Summary of Chapter 2 ... several paragraphs... 1/2 to 1 page

Based on the study presented in Chapter 2, the following gaps / needs are identified with potential to improve:

- question / gap / need
- ...
- question / gap / need

To answer these questions, the following starting points are proposed.

- Starting point 1:
- Starting point 2:
- Starting point 3:

Priebežná správa BP1/DP1 musí obsahovať:

1. Prehľad podobných riešení alebo produktov, s ktorými sa študent oboznámil pri štúdiu danej problematiky.
2. Špecifikáciu požiadaviek na vytvárané riešenie alebo produkt (Chapter 3).
3. Návrh
4. Vlastne zhodnotenie práce za príslušné obdobie (Chapter C - Section C.1).

Chapter 3

Objectives and Methodology

3.1 Objectives

Vybrané ciele pre navrhnuté riešenie alebo špecifikácia požiadaviek (funkčné a nefunkčné požiadavky) na naimplementovaný softvérový produkt.

- povinne pre DP a DiZP, ale aj BP môže
- max 2 pages for the whole chapter.

As presented in Chapter 2, the area of... is wide with many interesting unsolved problems. Therefore, the goals of this thesis are focused on selected objectives (maximum 3 objective) to... as follows:

- Objective 1:
- Objective 2:
- Objective 3:

3.2 Methodology

The following steps have been chosen for the design and development of ...

- Step 1:
- Step 2:
- ...
- Step n:

These steps are commitments to each other in the context of ... of the work.

Chapter 4

Design / Methods Description

Concrete header for the proposed design of the thesis's solution

This chapter can contain:

- Specification based on the Sections 2.6
- Design
- Technologies (can also be/moved in/to Chapter 5 Implementation)

Description how the design works (as much as possible from the below list)

- We need to separate our design from our concrete implementation
- Architecture, Block scheme, Event diagram, Sequence diagram, Algorithms, ...
- Text description of each blocks and their connections and/or dependencies
- Time complexity and memory complexity estimation for each execution step and the whole design. Communication complexity of the design (if any).

Figure 4.1 is about ... description of each blocks

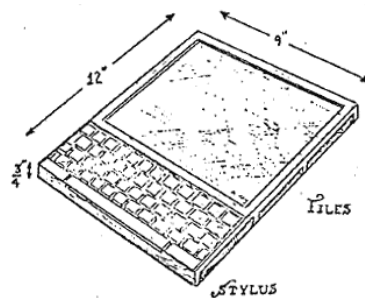


Figure 4.1: Architecture/Workflow

Chapter 5

Implementation

Concrete header for the implementation of the proposed design

The concrete realization of our proposed design

Concrete tools, software, solutions based on concrete dataset

Pseudocode of Algorithms, or Code snippets

$$\text{Bounce Rate} = \frac{\text{Single page sessions}}{\text{Total sessions}} \quad (5.1)$$

Chapter 6

Experiments and Evaluation

6.1 Setup

hardware,
software, libraries, versioning,
virtual environment description and setting,
datasets, NDA,
etc.

Machine	Remote	Google Colab	Kaggle	FIIT Cloud	Local
Python	3.9.19	3.10.12	-	-	3.10.12
PyTorch	-	-	-	-	-
TensorFlow	2.10	2.15	-	-	2.14
GPU (NVIDIA)	RTX xx	Tesla T4 15 GB	Tesla P100 16GB	-	-
CUDA	12.2	12.2	-	-	-
cuDNN	8.1.0	N/A	-	-	-
CUDA Toolkit	11.2.2	N/A	-	-	-
RAM	32/64 GB	51 GB	30 GB	32 GB	16 GB
CPU	8x cores Intel i7	-	Xeon 4x cores	4x cores	-
Storage	SSD	Google Driver	73 GB	128 GB	HDD

Table 6.1: Hardware, software and versioning used for realization

6.2 Case Study - Experiment 01

6.3 Case Study - Experiment 02

6.4 Case Study - Experiment 03

6.5 Summary

Main *key findings* of the work based on the Experiment results

- add ...
- ...

Discussion about various considerations based on the experiment results (if needed).

Regarding *limitations* of the work ...

- add ...
- ...

Chapter 7

Conclusion

7.1 Summary

Summary of the whole work including SOTA, design, implementations, experiments, evaluations, key findings ... 1 page

7.2 Future Work

1/2 page

Resumé

Chapter je bez cislovania, pouziva sa len **boldovanie textu** na zyzraznenie useku textu.
Bez Figure, Table, atd. ak treba tak odvolava sa na originalne Figure a Table.

10% v slovenčine pre tých, ktorí majú prácu v EN: preklad

EN	SK
Introduction	primerane strucne
SOTA	primerane strucne
SOTA - cast Summary and Starting Points	cele
Objective and Methodology	cele
Design, Implementation	primerane strucne
Experiments and Evaluation	primerane strucne
Experiments and Evaluation - cast Summary	cele
Conclusion and Future Work	cele

Table 7.1: Resumé

List of Abbreviations

ADE Average Displacement Error

AV autonomous vehicle

AVs autonomous vehicles

DBN dynamic Bayesian network

FDE Final Displacement Error

GMM Gaussian mixture model

GP Gaussian process

GPS global positioning system

HD map high-definition map

HMM hidden Markov model

IL imitation learning

IMU inertial measurement unit

IRL inverse reinforcement learning

LiDAR light detection and ranging

mAP mean Average Precision

MR Miss Rate

NLL Negative Log-Likelihood

RL reinforcement learning

SLAM simultaneous localization and mapping

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Appendix A

Description of Digital Submission

Thesis Evidence Number:

FIIT-XXXX-XXXXX

Obsah elektronického média je väčší ako 1GB a preto je odovzdaný vedúcemu záverečnej práce (doc. Ing. Giang Nguyen Thu, PhD.).

Name of submitted archive: DP_MenoPriezvisko.zip

The digital part included in the thesis contains the following files:

- Ubuntu OS: run in command line \$ `tree` to get a directory structure as below.
- Inspiration from <https://github.com/FIIT-ISA/cookiecutter-data-science>

ROOT // Root directory.

```
|
|— flatbuffers // This flatbuffers director
|— scheme
```



LICENSE	
Makefile	<- Makefile with commands like `make data` or `make train`
README.md	<- The top-level README for developers using this project.
data	
├── external	<- Data from third party sources.
├── interim	<- Intermediate data that has been transformed.
├── processed	<- The final, canonical data sets for modeling.
└── raw	<- The original, immutable data dump.
docs	<- A default Sphinx project; see sphinx-doc.org for details
models	<- Trained and serialized models, model predictions, or model summaries
notebooks	<- Jupyter notebooks. Naming convention is a number (for ordering), the creator's initials, and a short `-` delimited description, e.g. `1.0-jqp-initial-data-exploration`.
references	<- Data dictionaries, manuals, and all other explanatory materials.
reports	<- Generated analysis as HTML, PDF, LaTeX, etc.
└── figures	<- Generated graphics and figures to be used in reporting
requirements.txt	<- The requirements file for reproducing the analysis environment, e.g. generated with `pip freeze > requirements.txt`
setup.py	<- makes project pip installable (pip install -e .) so src can be imported
src	<- Source code for use in this project.
├── __init__.py	<- Makes src a Python module
├── data	<- Scripts to download or generate data
└── make_dataset.py	
├── features	<- Scripts to turn raw data into features for modeling
└── build_features.py	
├── models	<- Scripts to train models and then use trained models to make predictions
└── predict_model.py	
└── train_model.py	
└── visualization	<- Scripts to create exploratory and results oriented visualizations

Figure A.1: The structure of the ZIP file

Appendix B

Technical Documentation

B.1 Installation Manual

1. Clone the project from the Git repository (URL) OR unzip the ZIP file ...
2. Run `pip install -r requirements.txt` in the root folder of the project
3. Run `python -m spacy download en` to have the data corpus ...

B.2 User Manual / Používateľská príručka

1. Spustenie programu funguje pomocou hlavného Python súboru *main.py* príkazom `python main.py`.
2. Po spustení programu sa zobrazí grafické rozhranie programu v ktorom sa dá hneď na hlavnej obrazovke spustiť sumarizácia pre jeden vstupný súbor.
3. Tlačidlami v hornej časti obrazovky sa dá presúvať medzi funkcionalitami programu ako je viac súborová sumarizácia alebo zobrazovanie článkov. Na týchto obrazovkách sa tlačidlom *Back to main menu* dá vrátiť späť na hlavnú obrazovku.

Screenshot(s) používateľského grafického rozhrania (GUI) ako obrázok (Figure)

(Another variant without GUI:) The main objective of this work is to ... Several case studies were presented that examine the capabilities of ... to show the feasibility and achievability of ... Users in our work are considered to work directly with the code as described in Section B.1.

B.3 Developer Manual

if needed then add ...

Appendix C

Work Schedule

C.1 Work Schedule for BP1/DP1

Week	Plan
1. - 2.	...
3. - 4.	...
5. - 6.	...
7. - 8.	...
9. - 10.	...
11. - 12.	...

Some explanation text... (if needed) ... a few sentences

Self-Evaluation of Work Schedule for BP1/DP1: Vyjadrenie k plneniu plánu: some text ... one paragraph ... self-evaluation

(In the best case:) The BP/DP 1 work plan is considered satisfied and successful.

C.2 Work Schedule for BP2/DP2

Week	Plan
1. - 2.	...
3. - 4.	...
5. - 6.	...
7. - 8.	...
9. - 10.	...
11. - 12.	...

Some explanation text... (if needed) ... a few sentences

Self-Evaluation of Work Schedule for BP2/DP2: Vyjadrenie k plneniu plánu: some text ... one paragraph ... self-evaluation

(In the best case:) The BP/DP 2 work plan is considered satisfied and successful.

C.3 Work Schedule for DP3

Week	Plan
1. - 2.	...
3. - 4.	...
5. - 6.	...
7. - 8.	...
9. - 10.	...
11. - 12.	...

Some explanation text... (if needed) ... a few sentences

Self-Evaluation of Work Schedule for DP3: Vyjadrenie k plneniu plánu: some text ... one paragraph ... self-evaluation

(In the best case:) The DP 3 work plan is considered satisfied and successful.